

EXAMINATION QUESTION PAPER

Paper: Set A

Total Questions: 5

Total Marks: 9

Date: 09-01-2026

Q1. [1 marks] [Easy] [data structures]

Question: Which fundamental data structure organizes elements in a Last-In, First-Out (LIFO) manner?

- A) Queue
- B) Linked List
- C) Stack
- D) Tree

Correct Answer: C

Q2. [2 marks] [Medium] [data structures]

Question: Which data structure is most appropriate for implementing a Last-In, First-Out (LIFO) mechanism, characterized by constant-time complexity for both insertion (push) and deletion (pop) operations?

- A) Queue
- B) Singly Linked List
- C) Heap
- D) Stack

Correct Answer: D

Q3. [3 marks] [Hard] [data structures]

Question: Consider a scenario where an algorithm requires frequent, high-performance execution of two specific operations on a dynamic set of elements: 1) insertion of a new element, and 2) retrieval of the minimum element, with a worst-case time complexity requirement of $O(\log n)$ for both operations, where n is the number of elements. Which standard data structure is optimally suited to satisfy these precise complexity constraints?

- A) A standard Doubly Linked List
- B) A Fibonacci Heap
- C) A Binary Search Tree (BST) configured as a self-balancing structure (e.g., AVL or Red-Black Tree)
- D) A Min-Heap implemented using a contiguous array

Correct Answer: C

Q4. [1 marks] [Easy] [dbms]

Question: What is the primary function of a Database Management System (DBMS)?

- A) To manage and control access to data within a database.
- B) To exclusively format and display data for end-users.
- C) To develop and compile source code for software applications.
- D) To provide high-speed network connectivity for data transfer.

Correct Answer: A

Q5. [2 marks] [Medium] [dbms]

Question: Which ACID property guarantees that a transaction is treated as a single, indivisible unit, meaning it either completes entirely (commits) or has no effect at all (aborts) if an error occurs?

- A) Isolation

B) Consistency

C) Atomicity

D) Durability

Correct Answer: C